

WASP-12b

R-1722

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_190310 · created 2026-08-02T16:42:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458552.68738 +/- 0.00059 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.045
Transit depth (Rp/Rs)^2	2.22 +/- 1.34 [%]
Orbital Inclination (inc)	77.85 +/- 2.85
Airmass coefficient 1 (a1)	3.93 +/- 7.63
Airmass coefficient 2 (a2)	-0.55 +/- 1.68
Scatter in the residuals of the lightcurve fit is	616.28 %
Best Comparison Star	#1 - [574, 339]
Optimal Aperture	1.7
Optimal Annulus	3.75
Transit Duration (day)	0.11 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.045 vs 0.1178 (deviation 0.53σ)
Duration fit vs published	0.11 d vs 0.125 d (deviation 0.81σ)
Mid-time O–C	0.9 min
Depth SNR	2.5
Residual scatter	616.28 %

Light curve

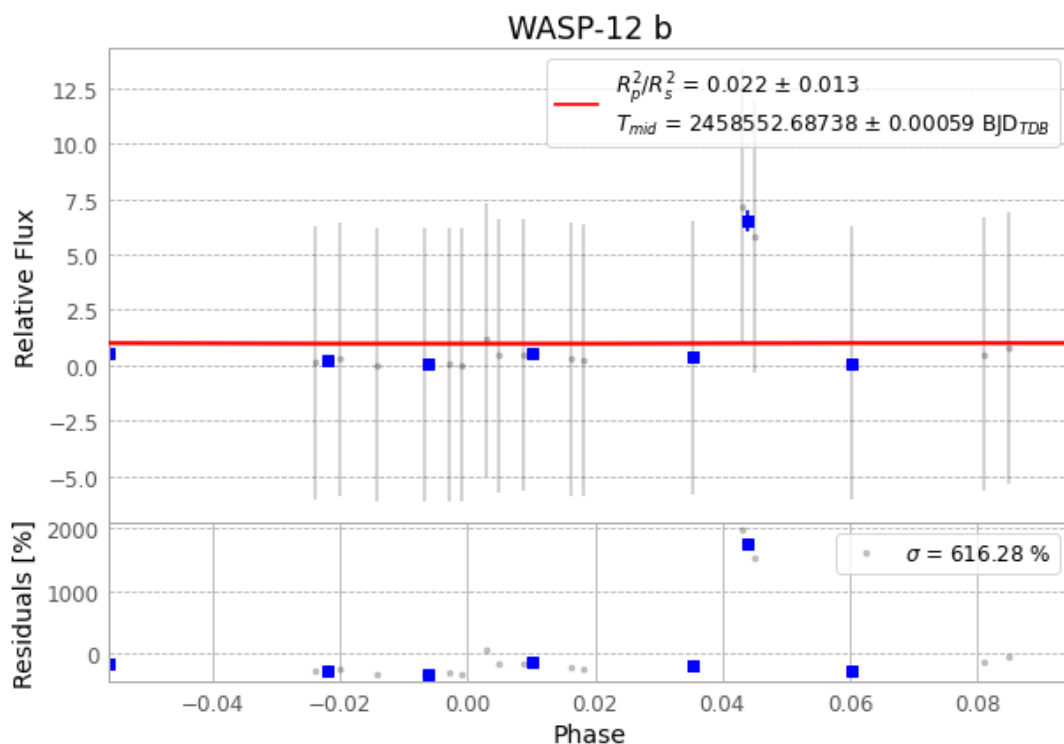


Figure 1 — FinalLightCurve_WASP-12 b_2019-03-09.png (SHA-256 4bbec33434e879b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [67, 406], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01ba4c562f21e809...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

93 FITS frames (61 MB), WASP-12190310022712.FITS → WASP-12190310071212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-190310.log (d9748aa328f9...), FinalLightCurve_WASP-12 b_2019-03-09.png (4bbec33434e8...), FinalLightCurve_WASP-12 b_2019-03-09.csv (55dc5a22ac62...)

Compute resources

Wall time 285.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:42Z.